

Supplementary information for MoLAG: monocular LED assignment via a graph neural network for multi-tracker surgical navigation

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S1 Scope

This supplement reports the complete architecture, loss and metric definitions, experimental settings, diagnostic results, and hyperparameter optimisation (HPO) record for MoLAG. All results come from the recorded experimental outputs.

S2 Reproducibility details

S2.1 Network and training

For N detections, $\mathcal{V} = \{1, \dots, N\}$, $\mathcal{E}_{\text{mp}} = \{(j \rightarrow i) : i, j \in \mathcal{V}, i \neq j\}$, and $\mathcal{E}_{\text{aff}} = \{\{i, j\} : i, j \in \mathcal{V}, i < j\}$. Thus, message passing is complete and directed, whereas affinity prediction represents each unordered pair once.

Each node begins with its normalised image coordinate $\mathbf{h}_i^{(0)} = \tilde{\mathbf{x}}_i$. For directed edge $i \rightarrow j$, the geometric attribute is

$$\mathbf{e}_{i \rightarrow j} = \left[\|\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i\|_2^2, \frac{\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i}{\max(\|\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i\|_2, \varepsilon_{\text{num}})} \right]. \quad (\text{S1})$$

Here $\varepsilon_{\text{num}} > 0$ prevents division by zero for coincident points. Each message-passing block uses the complete directed graph and applies

$$\mathbf{h}_i^{(\ell+1)} = \max_{j \neq i} \phi_{\theta_{\text{mp}}^{(\ell)}} \left([\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{j \rightarrow i}] \right), \quad (\text{S2})$$

where $\ell = 0, \dots, 6$ indexes the blocks, $\parallel$ denotes concatenation, the maximum is applied component-wise, and $\phi_{\theta_{\text{mp}}^{(\ell)}}$ is a two-layer ReLU network with parameters $\theta_{\text{mp}}^{(\ell)}$. The seven block widths are 128, 256, 512, 1,024, 2,048, 2,048, and 1,024. Concatenating the block outputs produces $128 + 256 + 512 + 1,024 + 2,048 + 2,048 + 1,024 = 7,040$ features per node. For every unordered pair of detections within a scene, the affinity head receives the Hadamard product and absolute difference of the two embeddings. Its two-layer multilayer perceptron contains 128 hidden units and an intermediate ReLU. The resulting affinity probabilities $p_{ij} = \sigma(z_{ij})$, where z_{ij} is the logit and σ the logistic sigmoid, weight $\hat{\mathcal{G}}_{\text{aff}} = (\mathcal{V}, \mathcal{E}_{\text{aff}}, p)$; they are scene predictions rather than model parameters.

Each full-fidelity run uses 5,000,000 generated scenes for ten epochs. An additional 10,000 scenes support in-training evaluation, and the checkpoint with the lowest such evaluation loss is retained. The physical batch size is 256 scenes. Four gradient-accumulation steps yield an effective batch size of 1,024. Training uses AdamW, a learning rate of 10^{-3} with cosine decay and 1 % warm-up, zero weight decay, bfloat16 arithmetic, and one NVIDIA RTX 5090 GPU. The implementation uses PyTorch 2.9.1 and PyTorch Geometric 2.7.0 [1].

S2.2 Tracker, camera, pose, and localisation-error model

The evaluated rigid tracker carries seven LEDs on an equilateral triangular body with a side length of 64 mm. Three LEDs occupy the vertices. Each side has three candidate positions. Two LEDs occupy side 0 and one occupies each of sides 1 and 2. For code $\mathbf{c} = (c_0, c_1, c_2)$, c_1 and c_2 select the occupied position on their respective sides, while c_0 identifies the one omitted position on side 0. This asymmetry produces 27 distinct configurations.

The generator uses an ideal pinhole camera with a 1440×1080 image, a 90° horizontal field of view, square pixels, and a centred principal point. Tracker centres are sampled with $x, y \in [-200, 200]$ mm and $z \in [100, 600]$ mm. A tracker’s local negative z axis first points towards the camera. It is then tilted by up to 85° about a random axis perpendicular to the viewing direction. Each scene contains one to ten trackers with distinct configuration codes.

Ray-triangle intersection tests model inter-tracker body occlusion. Projected LEDs behind the camera or outside the image are removed. In 80 % of scenes, LED dropout is applied independently with probability 0.4 while retaining at least three projected LEDs when possible. In 30 % of scenes, one to four spurious points are sampled within the image-space extent of the real detections. After mean centring and max-norm scaling, independent Gaussian coordinate noise with standard deviation 0.02 is applied in 80 % of scenes.

S2.3 Evaluation data

Calibration uses 5,000 scenes, with 500 per tracker count. The final test suite uses 50,000 scenes, with 5,000 per tracker count. The HPO validation suite contains 20,000 scenes. Training, in-training evaluation, calibration, HPO validation, and final test data are disjoint. Calibration and final testing otherwise use the same scene modifiers and synthetic distribution as training. For full-fidelity calibration, thresholds from 0.05 to 0.95 are evaluated in steps of 0.01. The threshold with the highest strict scene grouping is retained, with ties resolved in favour of the higher threshold.

S3 Complete loss definition

For a mini-batch of S scenes, $V_{s,t}$ denotes the visible nodes of tracker t , $\mathcal{T}_s$ the trackers in scene s , and B_s its spurious nodes. The affinity head assigns logit z_{ij} to unordered pair $\{i, j\}$. Every condition is formed within one scene before reduction across the batch. Let t_i be the ground-truth tracker identity of node i , using negative identities for spurious points. With $\mathbf{1}[\cdot]$ denoting the indicator function, the conventional binary pair target is

$$y_{ij} = \mathbf{1}[t_i = t_j \wedge t_i \geq 0]. \quad (\text{S3})$$

Binary cross-entropy treats these pair labels independently. The structured objective instead uses tracker identities to express the connectivity and separation conditions of the final partition.

S3.1 Shared reductions

For edge values v_a and non-negative handicaps η_a , indexed by $a \in A$, the weighted soft maximum is

$$\mathcal{M}_\beta(v, \eta; A) = \frac{1}{\beta} \left[\log \sum_{a \in A} \exp(\beta(v_a - \eta_a)) - \log \sum_{a \in A} \exp(-\beta\eta_a) \right]. \quad (\text{S4})$$

When at least one handicap is zero, it converges to $\max_{a \in A} (v_a - \eta_a)$ as $\beta \rightarrow \infty$. Finite β distributes gradient among several poor edges.

If scene s contains n_s applicable conditions with losses $q_{s,\kappa}$, the scene reduction is

$$\mathcal{R}_{\alpha,\rho}(q) = \frac{\sum_{s \in \mathcal{S}^*} \left(\frac{1}{n_s} \sum_{\kappa=1}^{n_s} q_{s,\kappa} \right) n_s^\alpha}{(1-\rho)S + \rho|\mathcal{S}^*|}, \quad \mathcal{S}^* = \{s : n_s > 0\}. \quad (\text{S5})$$

At $\alpha = 0$, each scene contributes its mean condition loss; at $\alpha = 1$, it contributes the sum. The parameter $\rho \in \{0, 1\}$ determines whether scenes without an applicable condition enter the outer mean. Connectivity and separation use separate powers α_c and α_s . Bridge reduction follows α_c , whereas attachment uses $\alpha = 0$.

S3.2 Tracker connectivity

Let $E_{s,t}^{\text{same}} = \{\{i, j\} : i, j \in V_{s,t}, i < j\}$. With connectivity, separation, and bridge margins m_c , m_s , and m_b , a maximum spanning tree $T_{s,t} \subseteq E_{s,t}^{\text{same}}$ is selected from the current logits, with the discrete selection detached from back-propagation. Using $\text{sp}(x) = \log(1 + \exp x)$,

$$\begin{aligned} C_{s,t} &= \mathcal{M}_\beta(\psi, \eta; E_{s,t}^{\text{same}}), & \psi_{ij} &= \text{sp}(m_c - z_{ij}), \\ \eta_{ij} &= \begin{cases} 0, & \{i, j\} \in T_{s,t}, \\ \delta, & \text{otherwise,} \end{cases} & \mathcal{L}_{\text{conn}} &= \mathcal{R}_{\alpha_c, \rho}(\{C_{s,t}\}). \end{aligned} \quad (\text{S6})$$

The discount δ limits the contribution of non-tree edges while preserving their gradient. No connectivity condition applies to a tracker with one observed node.

S3.3 Tracker-pair separation

For $t \neq u$, let $E_{s,t,u}^{\text{diff}} = \{\{i, j\} : i \in V_{s,t}, j \in V_{s,u}\}$. The strongest cross-tracker logit and separation loss are

$$d_{s,t,u} = \mathcal{M}_\beta(z, 0; E_{s,t,u}^{\text{diff}}), \quad \mathcal{L}_{\text{pair}} = \mathcal{R}_{\alpha_s, \rho}(\{\text{sp}(m_s + d_{s,t,u})\}_{t < u}). \quad (\text{S7})$$

Applying the soft maximum before the margin makes its hard limit the largest cross-tracker logit.

S3.4 Spurious attachments and bridges

For tracker t and spurious node b , define $E_{s,t,b}^{\text{att}} = \{\{i, b\} : i \in V_{s,t}\}$. Direct attachments are penalised by

$$\mathcal{L}_{\text{att}} = \mathcal{R}_{0, \rho}(\{\text{sp}(m_s + \mathcal{M}_\beta(z, 0; E_{s,t,b}^{\text{att}}))\}_{t, b}). \quad (\text{S8})$$

The fixed power $\alpha = 0$ prevents the independently sampled number of spurious points from changing scene weight.

For bridge risk, each tracker is contracted to one terminal. Set $W(t, b) = \max_{j \in V_{s,t}} z_{bj}$ and $W(b, b') = z_{bb'}$. The widest path through at least one spurious node is

$$\begin{aligned} b_{s,t,u} &= \max_{P: t \rightsquigarrow u, \text{int}(P) \subseteq B_s, |\text{int}(P)| \geq 1} \left(\min_{e \in P} W(e) \right), \\ \mathcal{L}_{\text{bridge}} &= \mathcal{R}_{\alpha_c, \rho}(\{\text{sp}(m_b + b_{s,t,u})\}_{t < u}). \end{aligned} \quad (\text{S9})$$

Here P is a path and $\text{int}(P)$ its internal nodes. A Floyd-Warshall widest-path closure computes $b_{s,t,u}$. If no such path exists, $b_{s,t,u} = -\infty$ and the pair contributes zero while remaining in the scene denominator.

S3.5 Embedding regularisation and complete objective

For real nodes $V_s^{\text{real}} = \bigcup_t V_{s,t}$, define normalised embeddings $\bar{\mathbf{h}}_i = \mathbf{h}_i / \|\mathbf{h}_i\|_2$ and $\text{sim}_{ij} = \bar{\mathbf{h}}_i^\top \bar{\mathbf{h}}_j / \tau_{\text{sc}}$, where τ_{sc} is the temperature. For anchors $\mathcal{A}_s = \{i : |P_s(i)| > 0\}$ with $P_s(i) = \{j \in V_s^{\text{real}} \setminus \{i\} : t_j = t_i\}$,

$$\mathcal{L}_{\text{supcon}} = \frac{1}{|\mathcal{S}_{\text{sc}}|} \sum_{s \in \mathcal{S}_{\text{sc}}} \frac{1}{|\mathcal{A}_s|} \sum_{i \in \mathcal{A}_s} \left[\log \sum_{j \in V_s^{\text{real}} \setminus \{i\}} e^{\text{sim}_{ij}} - \frac{1}{|P_s(i)|} \sum_{j \in P_s(i)} \text{sim}_{ij} \right], \quad (\text{S10})$$

where $\mathcal{S}_{\text{sc}} = \{s : |\mathcal{A}_s| > 0\}$. Spurious nodes and cross-scene pairs are excluded. The complete family is

$$\mathcal{L} = w_c \mathcal{L}_{\text{conn}} + w_s (\mathcal{L}_{\text{pair}} + \varepsilon_{\text{spur}} \mathcal{L}_{\text{att}}) + w_b \mathcal{L}_{\text{bridge}} + w_{sc} \mathcal{L}_{\text{supcon}}. \quad (\text{S11})$$

The combined full-fidelity setting uses $w_c = 1$, $w_s = 0.46$, $w_b = 0.25$, $m_c = m_s = 1$, $m_b = 0$, $\beta = 1$, $\delta = 3$, $\varepsilon_{\text{spur}} = 0.01$, $\alpha_c = \alpha_s = 0.5$, $\rho = 1$, and $w_{sc} = 0$.

S4 Complete metric definitions

Let $\mathcal{S}_{\text{test}}$ contain M_{test} scenes. For scene s , $\mathcal{V}_s$ is the node set, $V_{s,t}$ the nodes of tracker t , $\mathcal{T}_s$ the tracker set, B_s the spurious nodes, and $\hat{g}_s(i)$ the predicted component of node i ; $\mathbf{1}[\cdot]$ denotes the indicator function. Let τ be the calibrated affinity threshold. The retained affinity edges are $\hat{\mathcal{E}}_{\tau,s} = \{\{i, j\} : p_{ij} \geq \tau\}$.

S4.1 Strict grouping and recovery

The unsplit, unmerged, and strict-recovery indicators for tracker t are

$$\begin{aligned} u_{s,t} &= \mathbf{1}[|\{\hat{g}_s(i) : i \in V_{s,t}\}| = 1], \\ \nu_{s,t} &= \mathbf{1}[\nexists t' \neq t, i \in V_{s,t}, j \in V_{s,t'} : \hat{g}_s(i) = \hat{g}_s(j)], \\ r_{s,t} &= u_{s,t} \nu_{s,t}. \end{aligned} \quad (\text{S12})$$

Thus, strict recovery requires all visible LEDs of one tracker to share a component that contains no real LED from another tracker. Spurious detections may attach to one tracker without failure; they cause failure when they connect trackers. Strict scene grouping and all-tracker strict recovery are

$$g_s = \prod_{t \in \mathcal{T}_s} r_{s,t}, \quad A_{\text{group}} = \frac{1}{M_{\text{test}}} \sum_s g_s, \quad R_{\text{all}}^{\text{strict}} = \frac{\sum_s \sum_{t \in \mathcal{T}_s} r_{s,t}}{\sum_s |\mathcal{T}_s|}. \quad (\text{S13})$$

Removing every edge incident to a spurious node defines component assignment $\hat{g}_s^{\text{real}}$ and corresponding indicator $r_{s,t}^{\text{real}}$. Real-only grouping is $A_{\text{group}}^{\text{real}} = M_{\text{test}}^{-1} \sum_s \prod_{t \in \mathcal{T}_s} r_{s,t}^{\text{real}}$.

S4.2 Complete-tracker measures

Let $\mathcal{T}_s^{(7)} = \{t \in \mathcal{T}_s : |V_{s,t}| = 7\}$. The complete-tracker share and strict recovery rate are

$$Q_{\text{complete}} = \frac{\sum_s |\mathcal{T}_s^{(7)}|}{\sum_s |\mathcal{T}_s|}, \quad R_{\text{complete}}^{\text{strict}} = \frac{\sum_s \sum_{t \in \mathcal{T}_s^{(7)}} r_{s,t}}{\sum_s |\mathcal{T}_s^{(7)}|}. \quad (\text{S14})$$

For an unsplit tracker, let $\hat{g}_{s,t}$ be its single component. Extractability counts each component containing at least one unsplit complete tracker once:

$$E_{\text{complete}} = \frac{\sum_s \left| \left\{ \hat{g}_{s,t} : t \in \mathcal{T}_s^{(7)}, u_{s,t} = 1 \right\} \right|}{\sum_s |\mathcal{T}_s^{(7)}|}. \quad (\text{S15})$$

A split tracker contributes zero. If two complete trackers share a component, that component contributes once.

Fig. S1 illustrates the principal success and failure patterns used by the diagnostics below.

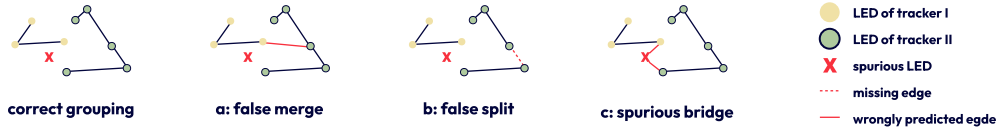


Fig. S1 Principal grouping outcomes. Node colour denotes the true tracker; nodes of tracker II also have a dark outline. The cross denotes a spurious detection. Dark edges are correct affinities; solid red edges create merges, while the missing connection marked by the dashed red line causes a split. A false merge can fail strict grouping while retaining an extractable complete tracker

S4.3 Edge and failure diagnostics

For pair label $y_{ij} = \mathbf{1}[i, j \notin B_s \wedge t_i = t_j]$ and prediction $\hat{y}_{ij} = \mathbf{1}[p_{ij} \geq \tau]$, aggregate true positives, false positives, and false negatives over edge set A :

$$P_A = \frac{\text{TP}_A}{\text{TP}_A + \text{FP}_A}, \quad R_A = \frac{\text{TP}_A}{\text{TP}_A + \text{FN}_A}, \quad F_{1,A} = \frac{2P_A R_A}{P_A + R_A}. \quad (\text{S16})$$

Here TP_A , FP_A , and FN_A are the corresponding aggregate counts. All-pair metrics use every unordered within-scene pair. Real-real metrics include only pairs of true LEDs.

Define the scene split and merge indicators $\xi_s^{\text{split}} = 1 - \prod_t u_{s,t}$ and $\xi_s^{\text{merge}} = 1 - \prod_t \nu_{s,t}$. Let $\xi_s^{\text{merge,real}}$ use components after removing spurious-incident edges, and let $\xi_s^{\text{bridge}} = \xi_s^{\text{merge}}(1 - \xi_s^{\text{merge,real}})$. The mutually exclusive outcome indicators are

$$\begin{aligned} I_s^{\text{correct}} &= (1 - \xi_s^{\text{split}})(1 - \xi_s^{\text{merge}}), & I_s^{\text{split}} &= \xi_s^{\text{split}}(1 - \xi_s^{\text{merge}}), \\ I_s^{\text{mixed}} &= \xi_s^{\text{split}}\xi_s^{\text{merge}}, & I_s^{\text{bridge}} &= (1 - \xi_s^{\text{split}})\xi_s^{\text{bridge}}, \\ I_s^{\text{merge}} &= (1 - \xi_s^{\text{split}})\xi_s^{\text{merge}}(1 - \xi_s^{\text{bridge}}). \end{aligned} \quad (\text{S17})$$

Each rate is $M_{\text{test}}^{-1} \sum_s I_s$. The broader spurious-bridge rate $M_{\text{test}}^{-1} \sum_s \xi_s^{\text{bridge}}$ also includes scenes with splits.

The direct spurious-attachment rate is

$$A_{\text{spur}} = \frac{\sum_s \sum_{b \in B_s} \mathbf{1}[\exists i \in \mathcal{V}_s \setminus B_s : \{b, i\} \in \hat{\mathcal{E}}_{\tau, s}]}{\sum_s |B_s|}. \quad (\text{S18})$$

It is descriptive rather than an error by itself.

S4.4 Statistical procedures

Strict scene grouping uses 95 % Wilson score intervals. Complete-tracker strict recovery and extractability use scene-clustered ratio intervals with a delta-method variance estimate. Full-fidelity checkpoint comparisons use paired McNemar tests on strict scene grouping and the scene event that every complete tracker is extractable; signed z statistics use a two-sided 5 % level. Each full-fidelity configuration was trained once, so these tests quantify finite-test-suite uncertainty rather than training variation.

Each Phase-B HPO arm comprises three runs and is summarised by its mean and sample standard deviation. The primary selected-candidate comparison also retains an earlier incumbent run and pools within-arm variance across four arms to form a two-sided t interval. Secondary comparisons use the three Phase-B runs per arm. All HPO inference is limited to 6 % training fidelity.

S5 Detailed full-fidelity results

S5.1 Loss configurations and paired comparisons

Table S1 reports the absolute result for each full-fidelity checkpoint. Every row uses one training run and the same frozen test suite. The combined scaled-conjunction checkpoint achieved the highest strict grouping, complete-tracker extractability, real-real edge F_1 , and the lowest spurious attachment rate among the six settings.

Table S1 Full-fidelity results. SG: strict scene grouping; Ext.: complete-tracker extractability; CSR: complete-tracker strict recovery; ASR: all-tracker strict recovery; F_1 : real-real edge F_1 ; SA: spurious-attachment rate. All entries after the threshold are percentages

Setting	Threshold	SG	Ext.	CSR	ASR	F_1	SA
Conjunction reference	0.71	55.078	88.699	79.393	74.946	90.967	75.012
Dense gradient	0.77	55.336	88.106	79.709	76.317	91.425	73.623
Reduced separation	0.88	54.482	88.631	78.864	74.584	91.025	73.957
Scene weighting	0.76	56.330	88.623	80.557	76.312	91.259	73.250
No SupCon	0.78	54.638	87.105	79.184	75.697	90.908	70.913
Combined scaled conjunction	0.84	56.426	89.642	80.360	76.204	92.339	68.020

Table S2 reports rate differences and paired tests against the conjunction reference. The extractability difference is the tracker-level rate change. Its paired test instead uses the scene-level event that every complete tracker in a scene is extractable. The tests account for the shared frozen scenes. They do not include variation between training runs and are therefore checkpoint-level comparisons.

Table S2 Full-fidelity checkpoint comparisons with the conjunction reference. Differences are percentage-point changes in strict grouping and tracker-level extractability. Signed McNemar z statistics use the corresponding binary scene outcomes on the common frozen test suite

Setting	Δ SG	z_{SG}	Δ Ext.	$z_{Ext.}$
Dense gradient	+0.258	+2.25	-0.593	-9.04
Reduced separation	-0.596	-6.13	-0.068	-1.85
Scene weighting	+1.252	+12.13	-0.075	-1.46
No SupCon	-0.440	-4.25	-1.594	-14.32
Combined scaled conjunction	+1.348	+12.58	+0.944	+5.03

Fig. S2 extends the main two-metric comparison. It shows how each setting changes the two strict-recovery rates and edge behaviour relative to the reference. The combined checkpoint has favourable differences in all displayed measures, with the largest change in spurious attachment.

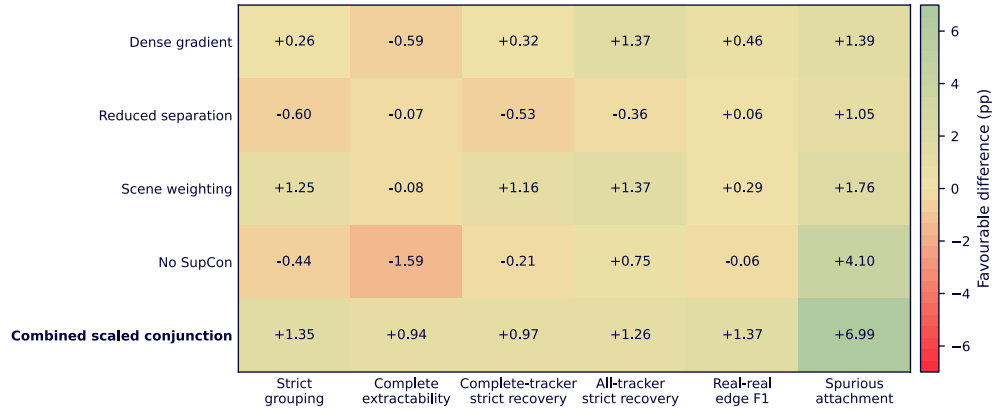


Fig. S2 Full-fidelity differences from the conjunction reference. Positive values are favourable. The sign of spurious attachment is reversed because a lower rate is better. Each row is one trained checkpoint evaluated on the common frozen test suite

S5.2 Diagnostic results

Table S3 reports the mutually exclusive outcome counts for the combined checkpoint. Merges and mixed errors account for most incorrect scenes. Only 66 scenes fail solely through a path of spurious detections.

Table S3 Mutually exclusive outcomes for the 50,000-scene frozen test suite

Outcome	Scenes	Rate (%)
Correct grouping	28,213	56.426
False split	1,276	2.552
False merge	10,511	21.022
Mixed split and merge	9,934	19.868
Spurious bridge only	66	0.132

The 27 tracker configuration codes occur at comparable frequency. All-tracker and complete-tracker strict recovery are also consistent across the codes (Fig. S3). Complete-tracker intervals are wider because complete trackers form the smaller subset.

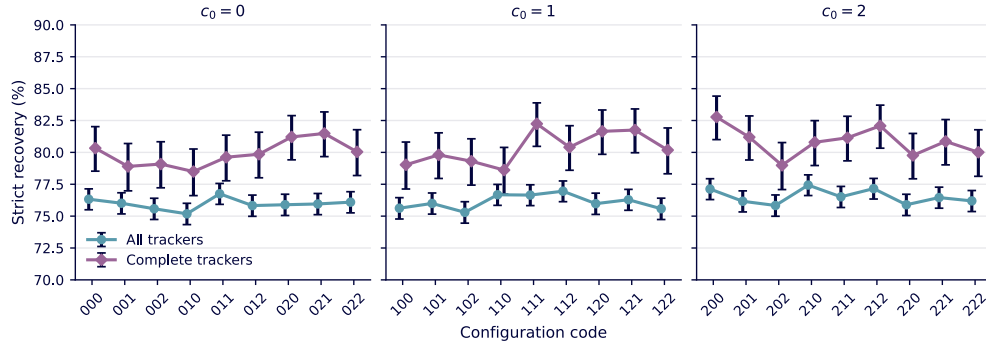


Fig. S3 All-tracker and complete-tracker strict recovery across the 27 configuration codes. A three-digit label identifies (c_0, c_1, c_2) , and panels group configurations by c_0 . Blue circles include all tracker instances. Purple diamonds include only complete seven-LED trackers. Whiskers are 95 % Wilson intervals

Performance by total detected-point count provides a complementary view of scene complexity (Fig. S4). Strict scene grouping falls as inputs grow. All-tracker strict recovery decreases more gradually, while complete-tracker extractability remains high across the observed range. Point count also changes with tracker count, visibility, and clutter, so this stratification is descriptive.

Complete-tracker strict recovery and extractability diverge with crowding because extractability accepts an intact tracker inside a merged component (Fig. S5).

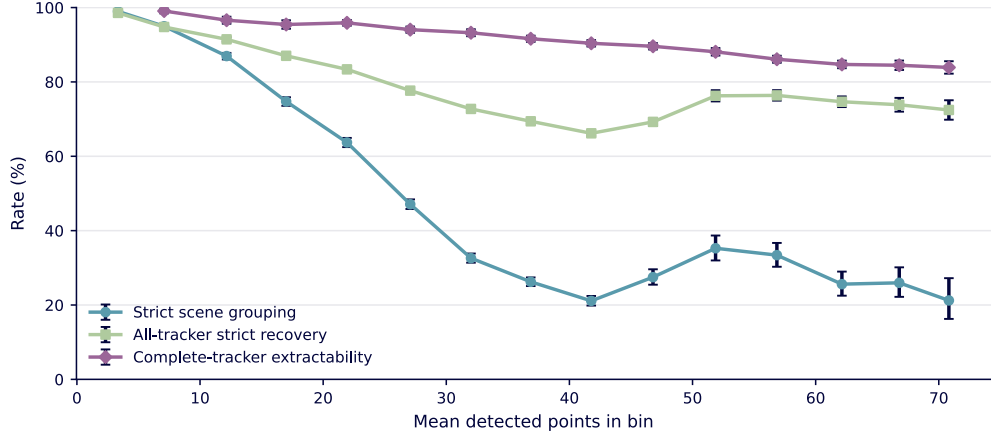


Fig. S4 Performance by total detected-point count for the combined configuration. Scenes are grouped into five-point bins, and each horizontal coordinate is the observed mean within its bin. Strict scene grouping uses Wilson intervals. All-tracker strict recovery and complete-tracker extractability use scene-clustered ratio intervals

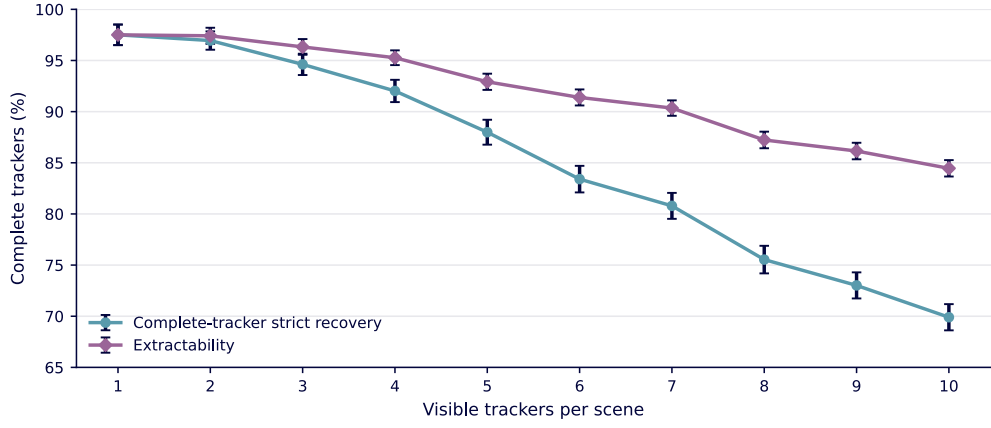


Fig. S5 Complete-tracker outcomes by visible tracker count. Strict recovery requires an unsplit and unmerged tracker; extractability also accepts an intact tracker in a merged component. Whiskers are 95 % scene-clustered ratio intervals

Strict recovery varies non-monotonically with visible LEDs and generally decreases with crowding (Figs. S6 and S7). Complete seven-LED trackers and sparse one- or two-LED cases have higher recovery than the intermediate visibility levels; false merges dominate throughout.

Strict grouping decreases with spurious detections, although spurious-bridge-only failures remain rare (Fig. S8).

Merges and mixed errors dominate incorrect partitions and increase with tracker count (Figs. S9 and S10).

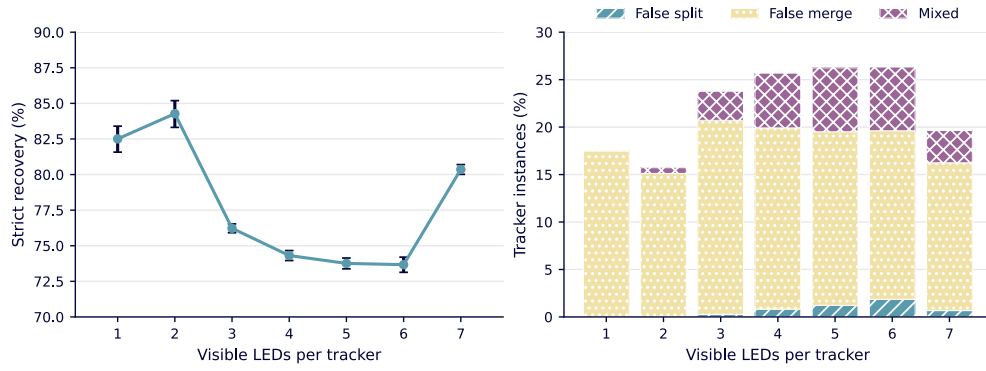


Fig. S6 Tracker-level outcomes by visible LEDs. The panels show strict recovery and mutually exclusive false-split, false-merge, and mixed-error rates

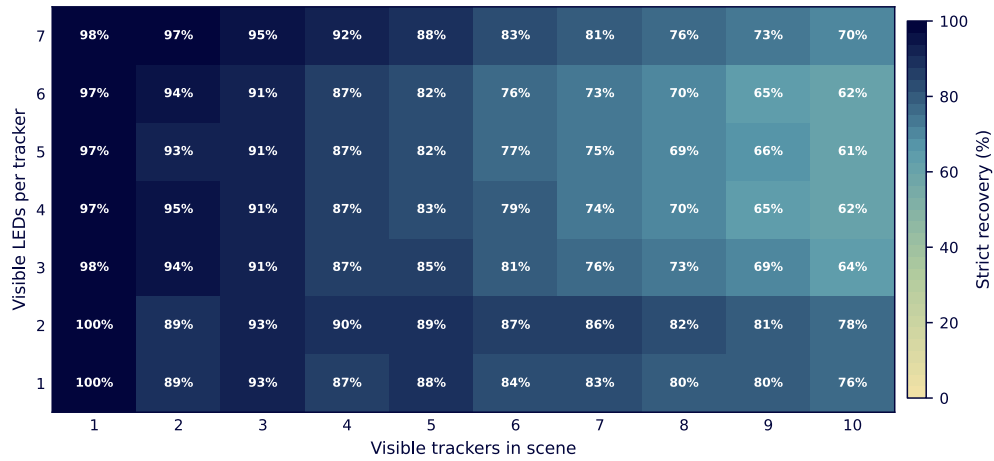


Fig. S7 Strict recovery by visible tracker count and visible LEDs per tracker. Cell labels give percentages

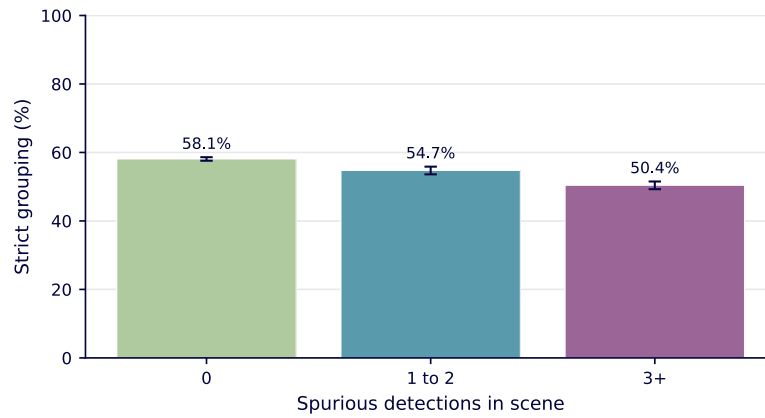


Fig. S8 Strict scene grouping by spurious-detection count. Labels give rates and whiskers show 95 % Wilson intervals

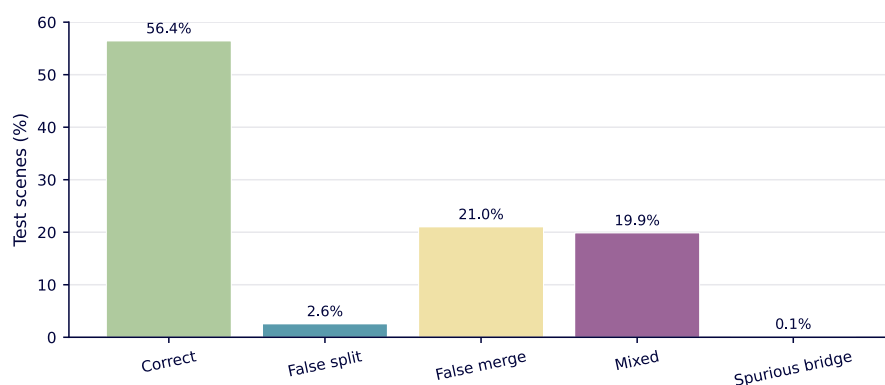


Fig. S9 Mutually exclusive outcomes for the 50,000 frozen scenes. Bar labels give percentages

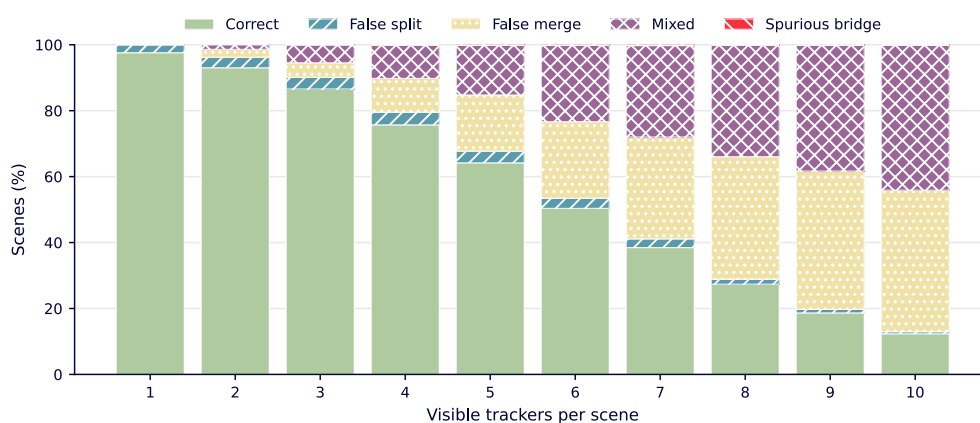


Fig. S10 Mutually exclusive scene outcomes by exact visible tracker count. Each bar represents 5,000 scenes and sums to 100 %

S6 Additional scope considerations

The synthetic distribution is a controlled stress model rather than a measured model of an operating room. In 80 % of scenes, dropout uses a pre-constraint per-LED probability of 0.4 and retains at least three projected LEDs when possible. Projection can still leave one or two LEDs. Physical occlusion is spatially structured by the camera, trackers, instruments, and anatomy, whereas the post-projection dropout is random. Without measured localisation data, the relative difficulty of these effects in deployment is unknown.

Spurious points occur in 30 % of scenes and are sampled uniformly inside the extent of real detections. This tests nearby clutter but not reflections or the appearance distribution of false detections. Independent Gaussian noise does not model camera distortion, detector thresholds, merged detections, motion, or calibration error. The ideal pinhole camera, stylised poses, distinct within-scene codes, triangular geometry, and removal of global position and scale further delimit the generalisation claim.

Each sample represents one independent frame. Temporal association, camera fusion, synchronisation, communication, and motion-dependent localisation are not evaluated. Runtime and pose performance also remain unmeasured. Extractability assumes that the downstream method can identify a complete tracker among extra detections, and a corresponding geometric procedure for the triangular tracker has not yet been tested. Recorded observations from several peer cameras are also needed to determine whether complementary views improve tracker availability when one view is incomplete.

S7 Exploratory HPO study

The HPO had three stages (Table S4). Phase 0 checked whether short runs reproduced the ordering of known full-fidelity configurations. Phase A searched the loss space at 6 % training fidelity. Phase B repeated selected settings and measured variation between training runs.

S7.1 Design and proxy check

Table S4 HPO stages and their roles

Stage	Completed work	Fidelity	Role
Phase 0	Seven known settings	3 %	Initial proxy check
Phase A	77 search settings	6 %	Candidate search
Phase B	Four arms, three runs each	6 %	Replication and noise estimate

Phase 0 did not establish rank fidelity. Seven known settings were evaluated; the rank comparison covered the six conjunction-family settings and excluded the

connectivity-only baseline. Their Spearman correlation with the full-fidelity ordering was -0.486 at 3 % fidelity. At 6 %, it was -0.600 with the single incumbent run and 0.086 after replacing that observation with the later repeat mean. These unstable estimates from six settings do not validate proxy rankings. The HPO is therefore interpreted through its direct Phase-B replication at 6 % fidelity.

Each Phase-A setting trained on 1,000,000 generated scenes for three epochs. Even-indexed validation scenes formed split A for threshold selection. Odd-indexed scenes formed split B for scoring. The objective was complete-tracker extractability on split B, with strict grouping of at least 0.4839 . Fifty-four of the 77 completed settings met this final floor.

S7.2 Search patterns and selected settings

Fig. S11 shows the objective and constraint for every completed search setting. The three ringed settings were retained for replication because they represented distinct parts of the high-extractability region. Trial 71 had the highest Phase-A objective and is therefore labelled the selected candidate. Replication tests the surrounding region rather than establishing a unique optimum.

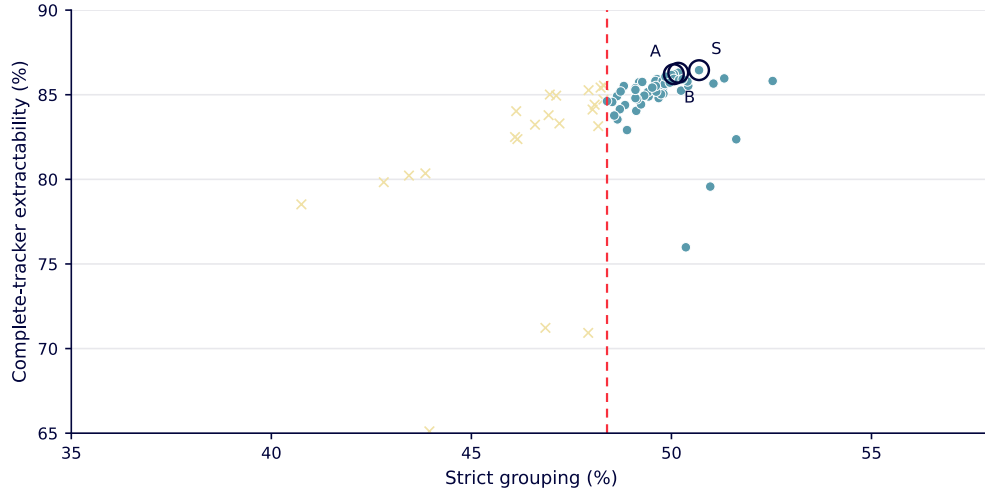


Fig. S11 Complete-tracker extractability and strict grouping for 77 completed search settings at 6 % fidelity. Blue circles meet the final strict-grouping floor of 48.39% , while yellow crosses fall below it. The dashed line marks the floor. Rings identify candidate A, candidate B, and the selected candidate

All 20 highest-scoring feasible settings used finite soft maximum aggregation. Table S5 summarises the other shared patterns. Higher-scoring feasible settings were associated with greater separation scene power and lower separation weights.

Table S6 reports the exact retained settings. Candidate labels are used in the figures and main text. Trial numbers link them to the complete Phase-A record. Each setting uses eligible-scene averaging and finite soft maximum aggregation. The non-tree discount is $\gamma_{nt} = \exp(-\beta\delta)$.

Table S5 Median parameters among the 20 highest- and 20 lowest-scoring feasible Phase-A settings

Parameter	Highest 20	Lowest 20
Connectivity scene power α_c	0.626	0.379
Separation scene power α_s	0.887	0.503
Separation weight w_s	0.285	0.569

Table S6 Exact parameters of the three settings retained for replication. A zero SupCon weight disables that term

Candidate	m_c	m_s	α_c	α_s	w_s	β	γ_{nt}	w_{sc}
A (trial 36)	1.93439	3.60479	0.66497	0.87918	0.28686	56.26569	0.03105	0.054485
B (trial 41)	0.50000	0.50000	0.73415	1.00000	0.45690	5.91172	0.00411	0
Selected (trial 71)	0.50000	4.00000	0.57319	0.87674	0.25780	3.11608	0.22098	0.009626

S7.3 Replication and measured training variation

The three retained candidates replicated within a narrow range and all exceeded the three-run incumbent mean (Table S7 and Fig. S12). Their similarity supports a high-performing region rather than a unique optimum.

Table S7 Complete-tracker extractability on split B for Phase-B runs at 6 % fidelity. Values are percentages. SD is the within-arm standard deviation in percentage points

Arm	Run 1	Run 2	Run 3	Mean	SD
Proxy incumbent	84.9060	84.3240	84.4862	84.5721	0.3004
Candidate A (trial 36)	86.0223	85.6788	86.1750	85.9587	0.2541
Candidate B (trial 41)	85.8029	86.1273	85.9555	85.9619	0.1623
Selected (trial 71)	86.4421	86.4708	85.9746	86.2958	0.2785

The primary comparison retains an earlier same-fidelity incumbent observation of 82.9119 %. The four-run incumbent mean is 84.1570 %. Pooling within-arm variation across the four replicated arms after retaining this observation yields $\sigma = 0.5359$ percentage points with 9 degrees of freedom. The selected candidate exceeds the incumbent by 2.1388 percentage points. The two-sided test yields $t = 5.226$, $p = 5.5 \times 10^{-4}$, and a 95 % interval of [1.213, 3.065] percentage points. The interval excludes zero after accounting for the measured between-run variation. This noise estimate and inference apply only to the 6 % fidelity design.

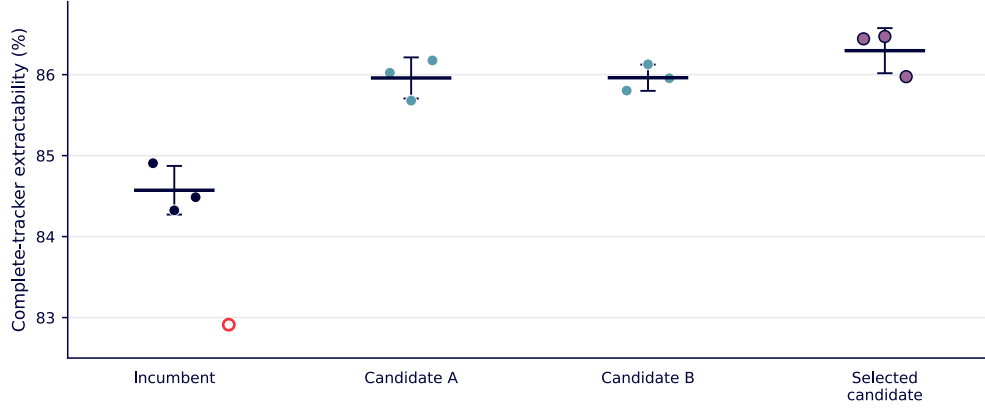


Fig. S12 Replicated complete-tracker extractability after 6 % fidelity training. Points show three runs per arm; bars and whiskers show the mean and sample standard deviation. The open point is the earlier incumbent observation retained in the primary comparison

Table S8 shows the broader operating-point change. The selected candidate exceeds the proxy incumbent in both complete-tracker measures. Strict scene grouping and all-tracker strict recovery remain similar. Higher real-real recall is accompanied by lower precision, lower edge F_1 , and more spurious attachment.

Table S8 Three-run sensitivity comparison of the selected candidate with the proxy-fidelity incumbent. Differences are percentage points. The reported p values use the pooled three-run analysis; the primary extractability inference is reported in the paragraph above

Metric	Incumbent (%)	Selected (%)	Diff.	Inference
Complete-tracker extractability	84.57	86.30	+1.72	Primary result above
Complete-tracker strict recovery	70.76	72.66	+1.91	$p = 0.005$
Strict scene grouping	49.91	50.40	+0.49	$p = 0.19$
All-tracker strict recovery	71.81	71.91	+0.11	$p = 0.77$
Real-real edge F_1	88.76	87.96	-0.80	$p = 10^{-4}$
Spurious attachment	70.40	78.30	+7.90	$p < 10^{-7}$

S7.4 Complete Phase-A record

Table S9 lists every completed search setting. The separation and SupCon weights are shown as compact search coordinates; the search also varied the parameters in Table S6.

Table S9: Complete Phase-A search record. Extractability is reported as a fraction; grouping is strict scene grouping on split B

Trial	Extractability	Grouping (%)	Threshold	Feasible	w_s	w_{sc}
0	0.840473	49.12	0.675	Yes	2.0000	0.03000

Trial	Extractability	Grouping (%)	Threshold	Feasible	w_s	w_{sc}
1	0.835416	48.65	0.750	Yes	2.0000	0.03000
2	0.831409	48.17	0.875	No	0.9000	0.03000
3	0.852400	50.24	0.700	Yes	1.0000	0.03000
4	0.844290	49.24	0.675	Yes	2.0000	0.00000
5	0.829119	48.89	0.850	Yes	0.4600	0.00000
6	0.848011	49.10	0.950	Yes	0.3641	0.03888
7	0.843908	48.84	0.950	Yes	0.3077	0.16542
8	0.858315	50.15	0.850	Yes	0.3409	0.11884
9	0.853831	49.79	0.625	Yes	0.3579	0.00000
10	0.759851	50.36	0.400	Yes	0.6480	0.00678
11	0.849442	47.12	0.950	No	0.2112	0.03156
12	0.712241	46.85	0.400	No	3.0000	0.00404
13	0.855357	48.31	0.950	No	0.1415	0.00000
14	0.849156	48.64	0.575	Yes	0.2221	0.00000
15	0.803549	43.85	0.950	No	0.5784	0.16152
16	0.850110	46.96	0.950	No	0.1053	0.00935
17	0.848965	49.43	0.400	Yes	0.8095	0.21193
18	0.709283	47.92	0.400	No	1.2731	0.00963
19	0.841427	48.71	0.775	Yes	2.0810	0.00000
20	0.795726	50.97	0.400	Yes	0.4899	0.00000
21	0.859651	51.32	0.850	Yes	0.3122	0.24968
22	0.845721	48.52	0.900	Yes	0.2318	0.00000
23	0.637535	42.20	0.400	No	3.0000	0.05899
24	0.847152	48.30	0.900	No	0.1502	0.07877
25	0.857361	49.19	0.525	Yes	0.1393	0.00000
26	0.798302	42.81	0.950	No	0.2327	0.00000
27	0.859269	49.64	0.475	Yes	0.2513	0.00000
28	0.651083	43.95	0.400	No	0.7549	0.01710
29	0.846198	48.39	0.925	Yes	0.0723	0.00000
30	0.858124	49.60	0.925	Yes	0.4700	0.17110
31	0.858220	49.78	0.525	Yes	0.2106	0.00000
32	0.851541	49.41	0.750	Yes	0.2203	0.00000
36	0.862990	50.17	0.825	Yes	0.2869	0.05449
37	0.840282	46.12	0.950	No	0.1364	0.00000
38	0.855166	48.81	0.750	Yes	0.1855	0.00000
39	0.858983	49.94	0.675	Yes	0.2514	0.00000
40	0.832268	46.59	0.950	No	0.0704	0.00930
41	0.862131	50.06	0.525	Yes	0.4569	0.00000
42	0.853926	49.10	0.750	Yes	0.2664	0.00442
43	0.823681	51.62	0.400	Yes	0.6477	0.00000
44	0.861654	50.00	0.575	Yes	0.3295	0.00000
45	0.851923	48.73	0.775	Yes	0.1166	0.00000
46	0.856311	49.83	0.475	Yes	0.6491	0.00000
47	0.857743	50.02	0.600	Yes	0.6134	0.16542
48	0.850587	49.80	0.825	Yes	0.6992	0.13797
49	0.848011	49.68	0.475	Yes	2.6459	0.23740
50	0.837706	48.57	0.600	Yes	3.0000	0.03023
51	0.860700	49.85	0.575	Yes	0.2402	0.00000
52	0.825017	46.09	0.575	No	0.0500	0.00000
53	0.849442	49.32	0.750	Yes	0.2052	0.00000
54	0.850301	49.74	0.825	Yes	0.1960	0.00000
55	0.844099	48.08	0.950	No	0.0926	0.03131
56	0.833031	47.20	0.925	No	0.5002	0.00000
57	0.847343	49.14	0.900	Yes	0.1711	0.08862
58	0.851827	49.62	0.850	Yes	0.2215	0.00000
59	0.841237	48.03	0.700	No	0.2220	0.00000
60	0.848011	49.10	0.450	Yes	0.7989	0.13176
61	0.855166	49.61	0.675	Yes	0.1571	0.00000
62	0.855262	50.42	0.650	Yes	1.3017	0.01452
63	0.823776	46.15	0.775	No	3.0000	0.00000
64	0.859365	50.06	0.850	Yes	0.2838	0.00000
65	0.837897	46.93	0.875	No	0.0500	0.00000

Trial	Extractability	Grouping (%)	Threshold	Feasible	w_s	w_{sc}
66	0.857838	50.40	0.950	Yes	0.2667	0.00000
67	0.852877	49.10	0.675	Yes	0.3250	0.00000
68	0.854212	49.52	0.550	Yes	0.1645	0.00000
69	0.856598	51.05	0.700	Yes	0.4205	0.00550
70	0.857552	49.27	0.950	Yes	0.3072	0.01219
71	0.864517	50.69	0.525	Yes	0.2578	0.00963
72	0.802214	43.44	0.950	No	0.0500	0.00000
73	0.852686	47.93	0.450	No	0.1541	0.00000
74	0.857170	49.96	0.600	Yes	0.1710	0.00000
75	0.858124	52.53	0.400	Yes	0.3275	0.00000
76	0.858697	50.19	0.725	Yes	0.1422	0.00000
77	0.785230	40.75	0.950	No	0.0500	0.02503
78	0.859174	50.28	0.775	Yes	0.3046	0.00000
79	0.854212	48.23	0.750	No	0.0936	0.00000

S7.5 Interpretive scope

Phase-A selection and Phase-B replication both use split B. Replication measures training variation but does not remove all selection optimism. The parity split is not exactly stratified. The feasibility floor was updated and then applied to the complete search record; one high-objective setting lies 0.0008 below it. Threshold search was bounded to 0.40 to 0.95. A fresh-split protocol was prepared but did not produce a completed result in this study.

The margin parameters have a near-gauge freedom. Their sampled values therefore do not establish their importance. Conditional parameters were sampled randomly when enabled, so they do not locate an optimum. Most importantly, no HPO-selected setting was trained with the full-fidelity recipe. The replicated result establishes a high-extractability operating region at 6% fidelity.

References

- [1] Fey M, Lenssen JE (2019) Fast graph representation learning with PyTorch Geometric. In: ICLR 2019 Workshop on Representation Learning on Graphs and Manifolds, URL <https://arxiv.org/abs/1903.02428>, [arXiv:1903.02428](https://arxiv.org/abs/1903.02428)